# STATEMENT OF WORK

## AI-Based Cloud Detection and Localization in Satellite Images

*Software Project Management Course Project*

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| **Document Title:** | Statement of Work (SOW) |
| **Project Title:** | AI-Based Cloud Detection and Localization in Satellite Images |
| **Project Type:** | Artificial Intelligence / Machine Learning / Image Processing |
| **Course:** | 458 – Software Project Management |
| **Institution:** | University of Karachi |
| **Course Instructor:** | Dr. Humera Azam |
| **Team / Group No.:** | Group 4 |
| **Team Size:** | 5 Members |
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*Note: Bracketed fields ([ ]) are placeholders to be completed by the project team and/or course instructor prior to formal submission and approval.*

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## 1. Project Title and Basic Information

| Field | Details |
|---|---|
| Project Title | AI-Based Cloud Detection and Localization in Satellite Images |
| Project Type | Artificial Intelligence / Machine Learning / Image Processing |
| Primary Development Platform | Python-based development environment |
| Core Output | An AI model and application that detects and highlights cloud regions in satellite images |
| Course | 458 – Software Project Management |
| Institution | University of Karachi |
| Course Instructor / Supervisor | Dr. Humera Azam |
| Team Size | 5 members |
| Estimated Duration | [One academic semester, e.g., 14–16 weeks — to be confirmed against the course calendar] |
| Document Version | 1.0 |
| Date Prepared | August 22, 2026 |

## 2. Introduction and Project Background

Optical satellite imagery is a valuable source of information for environmental monitoring, land-use analysis, disaster response, and scientific research. However, clouds frequently obscure the Earth's surface in such imagery, which reduces the usability of affected images and complicates downstream analysis. When large volumes of satellite images must be reviewed, manually identifying cloud-covered regions is time-consuming, inconsistent, and does not scale well.

This project proposes a software system that applies artificial intelligence (AI) and image-processing techniques to automatically detect the presence of clouds in satellite images and to localize, or highlight, the regions where clouds occur. The system is being developed as a Software Project Management (SPM) course project by a team of five BS Computer Science students, and is intended to demonstrate the full software project life cycle — from requirements definition through design, implementation, testing, and delivery — using a real-world-inspired AI/ML problem as the technical vehicle.

This project is inspired by research on AI-based cloud detection that also explores onboard (in-orbit) deployment on satellite hardware. For the purposes of this academic project, onboard deployment is explicitly excluded from scope. The team's focus is restricted to cloud detection, localization, visualization, and performance evaluation using standard, ground-based computing resources (personal computers and/or cloud-based notebook environments such as Google Colab).

This Statement of Work (SOW) is derived from, and remains fully consistent with, the approved Project Charter for this project. It does not introduce features, requirements, or deliverables beyond what is defined in the Project Charter.

## 3. Purpose of the Statement of Work

The purpose of this Statement of Work (SOW) is to define, in clear and actionable detail, the work required to deliver the AI-Based Cloud Detection and Localization in Satellite Images project. This document translates the objectives and scope established in the Project Charter into a structured description of deliverables, tasks, responsibilities, schedule, resources, and acceptance criteria that the project team, and the course instructor acting as project sponsor, can use to plan, execute, monitor, and evaluate the project.

Specifically, this SOW is intended to:

- Formally define the boundaries of the project (what is, and is not, included).
- Provide a detailed work breakdown that can be used as the basis for a Work Breakdown Structure (WBS) and Gantt chart.
- Assign responsibilities across the five-member project team.
- Establish the technology stack and resources required to execute the project.
- Document assumptions, constraints, risks, and mitigation strategies.
- Define the quality expectations, acceptance criteria, and completion criteria used to determine whether the project has been successfully delivered.
- Serve as a reference document for subsequent project management artifacts (WBS, schedule/Gantt chart, risk register, resource plan, and status reports).

*This SOW is a course deliverable prepared for academic purposes as part of a Software Project Management course. It does not represent a commercial contract, and all dates, durations, and resource figures are indicative placeholders unless stated otherwise.*

## 4. Project Scope

The scope of this project is limited to the design, development, testing, and demonstration of a software-based AI system for cloud detection and localization in satellite images, executed entirely on ground-based computing resources. The project does not involve the design, construction, or operation of any actual satellite or onboard hardware.

### 4.1 In-Scope Work

The following items are within the defined scope of this project:

- Selection and preparation of a suitable labeled satellite-image dataset.
- Image preprocessing to prepare data for model development.
- Training and testing of an AI model for cloud detection.
- Detection of the presence/absence of clouds in a given satellite image.
- Basic localization (highlighting) of detected cloud regions.
- Cloud-coverage estimation, where feasible within the selected model and dataset.
- Visualization of results (original vs. processed/annotated images).
- Performance evaluation of the trained model using appropriate metrics.
- Development of a simple user interface for uploading an image and viewing results.
- Preparation of project documentation, including the final SPM report and presentation.

### 4.2 Out-of-Scope Work

The following items are explicitly excluded from the scope of this project:

- Onboard (in-orbit) deployment of the model on satellite hardware.

- Design, construction, or procurement of actual satellite hardware.
- Satellite launching, operation, or mission control activities.
- Real-time, global-scale satellite image processing or production infrastructure.
- Any feature not directly related to cloud detection and localization (e.g., other atmospheric or land-cover classification tasks, unless explicitly approved as future work).

*Any request to add functionality beyond the items listed above must be treated as a scope-change request, evaluated for feasibility and timeline impact, and formally approved by the team and course instructor before being incorporated (see Section 13, Constraints, and Section 14, Risks and Mitigation).*

## 5. Project Objectives

The objectives of this project, as defined in the Project Charter, are as follows:

1. Collect a suitable labeled satellite-image dataset.
2. Preprocess and prepare images for model development.
3. Develop and train an AI model for cloud detection.
4. Evaluate the model using appropriate performance metrics.
5. Localize or highlight detected cloud regions.
6. Estimate cloud coverage where feasible.
7. Develop a simple interface for uploading an image and viewing results.
8. Document the system, experiments, results, limitations, and future work.

*These objectives directly govern the deliverables, work breakdown, and acceptance criteria defined later in this document. No additional objectives are introduced by this SOW.*

## 6. Project Requirements

### 6.1 Functional Requirements

Functional requirements describe what the system must do. Each requirement is tagged as Core (mandatory for project completion) or Conditional (required to the extent supported by the chosen model/dataset, consistent with the Project Charter's phrasing).

| ID | Requirement | Priority |
|---|---|---|
| FR-01 | The system shall accept a satellite image as input. | Core |
| FR-02 | The system shall preprocess the image before inference. | Core |
| FR-03 | The system shall determine whether cloud regions are present. | Core |
| FR-04 | The system shall highlight or localize detected cloud regions where supported by the implemented model. | Conditional |
| FR-05 | The system shall display the original and processed images. | Core |
| FR-06 | The system shall provide model performance results during evaluation. | Core |
| FR-07 | The system shall provide a simple user interface for demonstration. | Core |

### 6.2 Non-Functional Requirements

| ID | Requirement | Priority |
|---|---|---|
| NFR-01 | The application should be understandable and easy to operate. | Core |
| NFR-02 | The model and results should be reproducible using the documented dataset and procedure. | Core |
| NFR-03 | The system should provide reasonable processing time for demonstration images. | Core |
| NFR-04 | The project should maintain clear documentation of datasets, experiments, and limitations. | Core |

*"Conditional" indicates that the depth of implementation depends on model and dataset capability, consistent with the Project Charter's Minimum Viable Product (MVP) strategy described under Risk Mitigation (Section 14).*

## 7. Detailed Work Breakdown / Scope of Work

This section decomposes the project into eight sequential phases, aligned with the Proposed System Workflow defined in the Project Charter: Satellite Image → Preprocessing → AI Model → Cloud Detection → Cloud Localization → Result → Visualization. Each phase lists its objective, key tasks, inputs, and outputs / exit criteria. This breakdown is intended to serve as the basis for the project's formal Work Breakdown Structure (WBS) and Gantt chart.

### 7.1 Phase 1: Requirements and Literature Review

Establish a clear, agreed understanding of project requirements and survey existing approaches to cloud detection in satellite imagery to inform design decisions.

***Key Tasks***

- Review and finalize the Project Charter and this SOW with the team and instructor.
- Conduct a literature review of existing cloud-detection approaches (traditional image processing and AI/ML-based).
- Identify candidate datasets, model architectures, and evaluation metrics.
- Define functional and non-functional requirements (Section 6) and confirm project scope boundaries (Section 4).

***Inputs***

- Project Charter
- Course guidelines and grading rubric
- Publicly available research literature

***Outputs / Exit Criteria***

- Finalized requirements list
- Short literature review summary
- Shortlist of candidate datasets and model approaches

### 7.2 Phase 2: Dataset Selection and Preparation

Select a suitable labeled satellite-image dataset and organize it for use in model development.

***Key Tasks***

- Evaluate candidate open datasets for suitability, licensing, and label quality.
- Select the primary dataset (and, if useful, a supplementary dataset) for cloud detection.
- Verify sample labels/annotations for correctness and consistency.
- Partition the dataset into training, validation, and test sets.
- Document dataset source, structure, and any known limitations.

***Inputs***

- Candidate dataset shortlist from Phase 1
- Storage/compute environment (local or Google Colab)

***Outputs / Exit Criteria***

- Prepared and partitioned dataset
- Dataset documentation (source, size, labeling scheme, splits)

## 7.3 Phase 3: Image Preprocessing

Prepare raw satellite images for consistent, reliable use in model training and inference.

***Key Tasks***

- Implement preprocessing operations (e.g., resizing, normalization, color/band handling, noise reduction) using OpenCV/NumPy.
- Apply data augmentation techniques where appropriate to improve model robustness.
- Build a repeatable preprocessing pipeline that can be applied consistently to training data and to new demonstration images.
- Validate preprocessing output visually and programmatically.

***Inputs***

- Prepared dataset from Phase 2
- Python, OpenCV, NumPy environment

***Outputs / Exit Criteria***

- Documented, reusable preprocessing pipeline
- Preprocessed training/validation/test image sets

## 7.4 Phase 4: AI Model Development

Design and implement a baseline AI model architecture suitable for cloud detection (and, where feasible, localization).

***Key Tasks***

- Select a baseline model architecture appropriate for cloud detection (e.g., a CNN-based classifier and/or segmentation model, subject to team decision).
- Implement the model using TensorFlow or PyTorch.
- Define the training configuration (loss function, optimizer, hyperparameters).

- Establish a baseline (first working, if imperfect) version of the model.

***Inputs***

- Preprocessed dataset from Phase 3
- TensorFlow/PyTorch environment (local or Google Colab)

***Outputs / Exit Criteria***

- Baseline model implementation
- Documented model architecture and design decisions

## 7.5 Phase 5: Model Training and Evaluation

Train the model on the prepared dataset and evaluate its performance using appropriate, documented metrics.

***Key Tasks***

- Train the baseline model and iterate on preprocessing, hyperparameters, or architecture to improve performance.
- Use available GPU resources (e.g., Google Colab) to manage training time.
- Evaluate the model using appropriate metrics (e.g., accuracy, precision, recall, F1-score, and/or IoU if localization/segmentation is implemented).
- Record and compare experiment results across iterations.
- Select the final model version for use in the remainder of the project.

***Inputs***

- Baseline model from Phase 4
- Training/validation/test datasets

***Outputs / Exit Criteria***

- Trained model (final selected version)
- Performance evaluation report / experiment log

## 7.6 Phase 6: Cloud Localization and Visualization

Localize or highlight detected cloud regions within images and visualize results, consistent with what the selected model supports.

***Key Tasks***

- Implement cloud region localization/highlighting (e.g., bounding regions, masks, or heatmaps, depending on model type).
- Implement cloud-coverage estimation where feasible (e.g., percentage of image area classified as cloud).
- Build visualization routines (using Matplotlib/OpenCV) to display original vs. processed/annotated images.
- Validate localization output against test images and known limitations.

***Inputs***

- Trained model from Phase 5
- Test images

***Outputs / Exit Criteria***

- Cloud localization/highlighting module
- Visualization routines
- Documented limitations of the localization approach

## 7.7 Phase 7: User Interface and System Integration

Develop a simple, functional interface that allows a user to upload an image and view detection/localization results, and integrate all modules into a single working system.

***Key Tasks***

- Design and implement a simple interface using Streamlit or Flask.
- Integrate the preprocessing pipeline, trained model, and localization/visualization modules behind the interface.
- Display original image, processed/annotated image, and relevant performance/result information to the user.
- Conduct integration testing across all modules.

***Inputs***

- All modules from Phases 3–6

***Outputs / Exit Criteria***

- Working end-to-end demonstration application
- Integration test results

## 7.8 Phase 8: Testing, Documentation, and Final Demonstration

Verify system quality, finalize all documentation, and prepare the final demonstration and presentation for the course.

***Key Tasks***

- Perform functional and non-functional testing against the requirements in Section 6.
- Test the system using previously unseen test images.
- Finalize project documentation, including datasets used, experiments conducted, results, limitations, and future work.
- Prepare the final SPM report and presentation slides.
- Conduct the final demonstration to the course instructor / evaluators.

***Inputs***

- Integrated system from Phase 7
- Test image set (unseen during training)

***Outputs / Exit Criteria***

- Tested, working system
- Final SPM report
- Final presentation
- Demonstration results

## 8. Project Deliverables

The following table lists the major deliverables of the project, consistent with the Project Charter. Each deliverable is classified as Core (required for project completion) or Optional/Stretch (included where time and model capability permit).

| ID | Deliverable | Description | Related Phase(s) | Classification |
|---|---|---|---|---|
| D1 | Prepared Dataset & Preprocessing Pipeline | A labeled dataset partitioned into training/validation/test sets, with a documented, reusable preprocessing pipeline. | Phase 2–3 | Core |
| D2 | Trained AI Model | A trained cloud-detection model with documented architecture and training configuration. | Phase 4–5 | Core |
| D3 | Cloud Detection Module | Software module that determines the presence/absence of clouds in a given input image. | Phase 4–5 | Core |
| D4 | Cloud Localization & Visualization Module | Module that highlights/localizes detected cloud regions and visualizes original vs. processed images; includes coverage estimation where feasible. | Phase 6 | Core / Conditional |
| D5 | Performance Evaluation Report | Documented evaluation of model performance using appropriate metrics, including experiment comparisons. | Phase 5 | Core |
| D6 | Working Demonstration Interface | A simple Streamlit/Flask interface allowing image upload and result viewing. | Phase 7 | Core |
| D7 | Final SPM Report & Presentation | Comprehensive documentation of the system, experiments, results, limitations, and future work, plus final presentation slides. | Phase 8 | Core |

## 9. Project Milestones and Schedule

The table below presents the project's major milestones, mapped to the eight work-breakdown phases defined in Section 7. Timeframes are expressed as indicative week ranges within a typical academic semester and must be aligned with the actual course calendar and instructor-defined checkpoints before being finalized in the project schedule / Gantt chart.

| ID | Milestone | Associated WBS Phase(s) | Indicative Timeframe | Key Exit Deliverable |
|---|---|---|---|---|
| M1 | Requirements, literature review, and dataset selection completed | Phase 1 | Weeks 1–2 | Finalized requirements; selected dataset |
| M2 | Dataset preparation and preprocessing completed | Phase 2–3 | Weeks 3–4 | Prepared dataset; preprocessing pipeline (D1) |
| M3 | Baseline model developed | Phase 4 | Weeks 5–6 | Baseline model implementation |
| M4 | Model training and evaluation completed | Phase 5 | Weeks 7–9 | Trained model (D2); evaluation report (D5) |

| | | | | |
|---|---|---|---|---|
| M5 | Cloud localization and visualization completed | Phase 6 | Weeks 10–11 | Localization/visualization module (D4) |
| M6 | Interface development and integration completed | Phase 7 | Weeks 12–13 | Working demonstration application (D6) |
| M7 | Testing, documentation, and final demonstration completed | Phase 8 | Weeks 14–16 | Final report, presentation, demonstration (D7) |

*Weeks are indicative planning placeholders (not fixed calendar dates) and are provided to support later development of a detailed schedule / Gantt chart. Actual dates must be confirmed against the course syllabus.*

## 10. Roles and Responsibilities

The project team consists of five members. Consistent with the Project Charter, responsibilities are organized into five role areas below. Exact member-to-role assignment is to be confirmed by the team at project kickoff.

| Role | Assigned To | Key Responsibilities |
|---|---|---|
| Project Manager / QA Lead | [Team Member Name] | Overall project coordination; requirement tracking; schedule and scope management; quality assurance; integration oversight; liaison with course instructor; risk monitoring. |
| Dataset & Preprocessing Lead | [Team Member Name] | Dataset collection, labeling/verification, and preprocessing; maintaining data documentation and reproducibility. |
| AI Model Development Lead | [Team Member Name] | AI model design, implementation, and training; hyperparameter tuning; managing compute resources (e.g., Google Colab). |
| Cloud Localization & Evaluation Lead | [Team Member Name] | Cloud localization and image-processing logic; performance evaluation and metrics; experiment logging. |
| Interface & Integration Lead | [Team Member Name] | User interface development (Streamlit/Flask); integration of modules; testing; documentation support. |

*All team members are expected to contribute to documentation, testing, and the final report/presentation regardless of primary role, consistent with standard academic team-project practice.*

## 11. Technology and Resource Requirements

### 11.1 Technology Stack

| Category | Technology | Purpose |
|---|---|---|
| Programming Language | Python | Primary development language for the entire system. |
| Image Processing | OpenCV | Image loading, preprocessing, and manipulation. |
| Numerical Computing | NumPy | Array and numerical operations supporting preprocessing and model logic. |
| Deep Learning Framework | TensorFlow or PyTorch | Model design, training, and inference. |
| Development Environment | Jupyter Notebook / Google Colab | Experimentation, model training (including free-tier GPU access via Colab), and iterative development. |
| Visualization | Matplotlib | Visualization of images, results, and evaluation metrics/charts. |

| | | |
|---|---|---|
| Interface Framework | Streamlit or Flask | Simple user interface for uploading images and viewing results. |

### 11.2 Other Resource Requirements

- Computing hardware: team members' personal computers, supplemented by cloud-based notebook environments (e.g., Google Colab) for GPU-accelerated training.
- Dataset access: a suitable open-source, labeled satellite-image dataset with appropriate licensing for academic use.
- Storage: sufficient local/cloud storage for datasets, trained model files, and experiment logs.
- Collaboration and version control tools (e.g., Git/GitHub or equivalent) to support team collaboration — recommended as good practice, though not mandated by the Project Charter.
- Course-provided resources: instructor guidance, grading rubric, and any institution-provided compute resources, where available.

*Any paid or licensed tools/datasets are out of scope unless separately approved, given the academic, no-budget nature of this project (see Section 13, Constraints).*

## 12. Assumptions

The planning and execution of this project rest on the following assumptions:

- A suitable labeled satellite-image dataset is publicly available and licensed for academic/research use.
- All five team members remain available and actively contribute throughout the project duration.
- Team members have, or will acquire during Phase 1, sufficient foundational knowledge of Python, image processing, and machine learning concepts.
- Free or low-cost computing resources (team members' personal computers and Google Colab's free tier) are sufficient to train and evaluate the model within the project timeline.
- The course instructor is available for periodic guidance, milestone review, and final evaluation.
- The project timeline follows the institution's standard academic semester calendar.
- Reliable internet access is available for dataset download, literature research, and use of cloud-based notebook environments.
- Model performance is evaluated on a demonstration basis and is not required to meet production-grade or operational satellite-mission standards.

## 13. Constraints

The project is subject to the following constraints:

- Fixed academic timeline: the project must be completed within a single academic semester, alongside coursework in other subjects.
- Limited budget: no funding is available for paid datasets, paid cloud compute, or specialized hardware; the project relies on free/open resources.
- Limited computing resources: free-tier GPU quotas (e.g., Google Colab) impose limits on training time and experiment volume.
- Team experience: team members are students who may have limited prior professional experience with large-scale AI/ML systems.

- Dataset availability and quality: the project is constrained by what labeled, cloud-annotated satellite datasets are publicly and freely available.
- Academic scope: the deliverable is a demonstration-level system intended for course evaluation, not a production or operational system.
- Fixed scope for core requirements: as agreed in the Project Charter's risk mitigation strategy, core requirements are frozen once approved, and any additional features are treated as future work rather than in-project scope changes.

## 14. Risks and Mitigation

The following risks, identified in the Project Charter, are carried forward into this SOW along with a preliminary qualitative likelihood/impact assessment (to be refined in the project's risk register) and the corresponding mitigation strategy.

| ID | Risk | Likelihood | Impact | Mitigation Strategy |
|---|---|---|---|---|
| R1 | Poor dataset quality | Medium | High | Use a suitable, well-established labeled open dataset and verify a sample of labels for correctness before committing to it. |
| R2 | Low model accuracy | Medium | High | Improve preprocessing, adjust training settings/hyperparameters, or select an alternative model architecture if initial accuracy is insufficient. |
| R3 | Long training time | Medium | Medium | Use available GPU resources (e.g., Google Colab) and manage dataset/model size to fit within compute and schedule constraints. |
| R4 | Difficulty in cloud localization | High | Medium | Treat cloud detection/classification as the Minimum Viable Product (MVP); treat localization as a subsequent milestone that is implemented to the extent supported by the chosen model. |
| R5 | Scope expansion (scope creep) | Medium | High | Freeze core requirements after approval; treat any additional/advanced features as documented future work rather than in-scope changes. |

*Likelihood and Impact ratings (Low/Medium/High) reflect the project team's preliminary planning assessment and are intended to support prioritization within the project's risk management process; they should be reviewed and updated periodically as the project progresses.*

## 15. Quality and Acceptance Criteria

### 15.1 Quality Assurance Approach

Quality will be managed throughout the project through: periodic code review among team members; incremental testing of each module (preprocessing, model, localization, interface) as it is completed; validation of results against held-out test images; and maintenance of clear, versioned documentation of datasets, experiments, and known limitations, consistent with the non-functional requirements in Section 6.2.

### 15.2 Acceptance Criteria

The following criteria, aligned with the Success Criteria defined in the Project Charter, will be used to assess whether each area of the system is acceptable for final delivery.

| ID | Related Requirement / Objective | Acceptance Criterion | Measure |
|---|---|---|---|
| AC-01 | FR-01, FR-02 | The system accepts a satellite image as input and applies preprocessing without errors. | Pass/Fail on test images |
| AC-02 | FR-03 | The trained model can detect cloud presence with measurable performance on unseen test images. | Accuracy / Precision / Recall / F1 reported |
| AC-03 | FR-04 | Cloud regions are visualized or localized within the boundaries of the selected project scope. | Visual/qualitative review + IoU or coverage estimate where applicable |
| AC-04 | FR-05, FR-06 | The system displays original and processed images and reports model performance results. | Demonstration walkthrough |
| AC-05 | FR-07, NFR-01 | The demonstration interface is understandable and operable by someone unfamiliar with the codebase. | Usability walkthrough / peer test |
| AC-06 | NFR-02, NFR-04 | Results are reproducible using the documented dataset and procedure, and documentation covers experiments, results, and limitations. | Documentation review |
| AC-07 | NFR-03 | Processing time for a single demonstration image is reasonable for a live demonstration. | Timed demonstration run |

## 16. Project Completion Criteria

The project will be considered complete when all of the following conditions, drawn directly from the Project Charter's Success Criteria, have been met and verified:

- A working system can accept satellite images as input.
- The trained model can detect cloud presence with measurable, documented performance.
- Cloud regions can be visualized or localized within the boundaries of the defined project scope.
- The system has been demonstrated successfully using previously unseen test images.
- All Core deliverables (D1–D7, Section 8) have been completed, reviewed, and accepted.
- Project documentation includes experiments conducted, evaluation metrics, limitations, and conclusions.
- The final SPM report and presentation have been submitted and evaluated by the course instructor.
- Formal sign-off has been obtained from the project team and course instructor (Section 17).

## 17. Approval / Sign-Off Section

By signing below, the project team members and course instructor acknowledge that they have reviewed this Statement of Work and agree that it accurately reflects the scope, deliverables, schedule, and responsibilities for this project, consistent with the approved Project Charter.

| Name | Role | Signature | Date |
|---|---|---|---|
| Tayyaba | Project Manager / QA Lead | | |
| Ayesha Mushtaq | Dataset & Preprocessing Lead | | |
| Fatima Fahad | AI Model Development Lead | | |
| Meerab Shafiq | Cloud Localization & Evaluation Lead | | |
| Samra Syed | Interface & Integration Lead | | |
| Dr. Humera Azam | Course Instructor / Project Sponsor | | |

*This section is a placeholder for formal, in-class or institution-defined approval procedures. Signature and date fields are to be completed at the time of formal review.*